(KTXFI1EBNF) Physics I. Lecture

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Current Updates and Issues

- Thank you for submitting your first homework assignment. I have graded the papers and provided the solutions. Three students did not submit their assignments; therefore, I must deny their signatures.
- I found errors in the material. Specifically, the direction of the force vectors and their starting points were not always correct. I have corrected these mistakes.
- The dates of the exams are not final. Let me know if you need more time or a different schedule.
- Future: we will finish Thermodynamics during this lecture. I will need more time for the practical part of this field. Then you will receive your 2nd assignment.
- ~~Electrodynamics~~ Electrostatics: no previous material available for lectures. Some mid-term exam exercises are available $\rightarrow$ We will practice them, and I will use my creativity.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** *Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.*
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Kinetic gas theory I

- Interpretation of the main thermodynamic quantities based on kinetic gas theory
- Pressure: $p = \frac{1}{3}\rho\overline{v^2}$ where $\overline{v^2}$ is the mean of v^2.
- Ideal gas equation of state:

$$pV = \frac{m}{M}RT$$

  $m = N \cdot m_0$, where m: mass of the gas, and m_0: mass of one particle
  $M = N_A \cdot m_0$, where M is the molar mass
- **Definition (Avogadro's number):**

$$N_A = 6 \cdot 10^{23}$$

Kinetic gas theory II

- According to these:

$$pV = \frac{Nm_0}{N_A m_0} RT = N \frac{R}{N_A} T$$

- **Definition (Boltzmann's constant):**

$$k = \frac{R}{N_A}$$

- Hence,

$$pV = NkT$$

Kinetic gas theory III

- The interpretation of temperature

$$
\begin{aligned}
p = \frac{1}{3}\rho\overline{v^2} &= \frac{1}{3}\frac{Nm_0}{V}\overline{v^2}\cdot V \\
pV &= \frac{1}{3}Nm_0\overline{v^2}
\end{aligned}
$$

- From previously: $pV = NkT$. Therefore,

$$
\begin{aligned}
\frac{1}{3}Nm_0\overline{v^2} &= NkT \quad : N \\
\frac{1}{3}m_0\overline{v^2} &= kT \quad \cdot 3 \\
m_0\overline{v^2} &= 3kT \quad : 2 \\
\frac{1}{2}m_0\overline{v^2} &= \frac{3}{2}kT
\end{aligned}
$$

Kinetic gas theory IV

- Equipartition theorem: each degree of freedom of a gas molecule has the same amount of kinetic energy, which is equal to $\frac{1}{2}kT$.
- $\frac{1}{2}m_0\overline{v^2}$ can be divided into 3 parts: according to the x, y, z coordinates. Hence,

$$\frac{3}{2}m_0\overline{v_i^2} = \frac{3}{2}kT \implies \frac{1}{2}m_0\overline{v_i^2} = \frac{1}{2}kT$$

- If the number of degrees of freedom is f:

$$\frac{1}{2}m_0\overline{v_i^2} = \frac{f}{2}kT$$

Kinetic gas theory V

- Internal energy of ideal gas:

$$U = N\frac{1}{2}m_0\overline{v^2} = N\frac{f}{2}kT$$

- Specific heat and molar heat:

$$U = c_v mT = N\frac{f}{2}kT = \frac{f}{2}\frac{m}{M}RT$$

Kinetic gas theory VI

- From the equation the c_v can be expressed:

$$
\begin{aligned}
c_v m T &= \frac{f}{2}\frac{m}{M}RT \\
c_v &= \frac{f}{2}\frac{R}{M} = \frac{f}{2}R_i \\
C_V &= c_v M = \frac{f}{2}R \\
c_p &= c_v + \frac{R}{M} = \frac{R}{M}(\frac{f}{2}+1) \\
C_p &= C_V + R = (\frac{f+2}{2})R
\end{aligned}
$$

Kinetic gas theory VII

- Boltzmann's definition of entropy:

$$S = k \cdot \ln w,$$

where w: number of microstates associated with a macrostate (thermodynamic probability), and k is Boltzmann-constant.

- The change in entropy according to Boltzmann's interpretation:

$$\Delta S = k \cdot \ln \frac{w_2}{w_1}.$$

- The change in entropy for ideal gas state changes:

$$\Delta S = \int_{T_1}^{T_2} \frac{c_v m \, dT}{T} + \int_{V_1}^{V_2} \frac{mR_i}{V} \, dV = c_v m \ln \frac{T_2}{T_1} + mR_i \ln \frac{V_2}{V_1}$$

Kinetic gas theory VIII

Thermodynamic distributions I

- Spatial distribution:

$$n_i = n_0 \cdot e^{\frac{-W_i}{kT}},$$

where n_i: number of particles in the i-th state, n_0: reference number of particles, usually representing the number in the most probable or ground state, W_i: the energy of the i-th state.

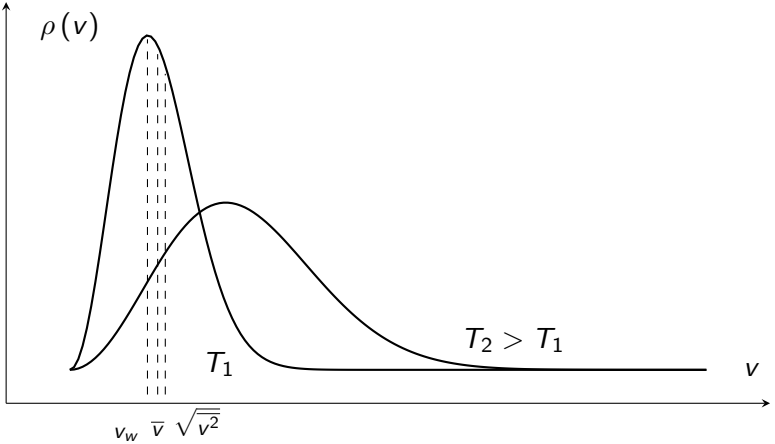
- Velocity distribution:

$$\rho(v) = \frac{dn}{dv} = 4\pi \left(\frac{m_0}{2\pi kT}\right)^{3/2} v^2 e^{\frac{-m_0 v^2}{2kT}},$$

where $\rho(v)$: the probability density function of the velocity distribution, $\frac{dn}{dv}$: the velocity density, showing how particles are distributed across different velocities. m is the mass of the particles. v is the velocity of the particles.

Thermodynamic distributions II

- Velocity Distributions at Different Temperatures



Thermodynamic distributions III

- Energy distribution: Maxwell-Boltzmann statistics

$$\rho(W) = \frac{dn}{dW} = \frac{8\pi}{m_0}\left(\frac{m_0}{w\pi kT}\right)^{3/2} We^{-\frac{W}{kT}}$$

- Binomial distribution: distribution of a population when N, the population size, is not too big

$$W_n = \frac{N!}{n_1! \cdot (N-n_1)!} p^{n_1} \cdot q^{(N-n_1)},$$

where n_1: number of samples of one type, p: probability of occurrence of n_1 sample, and q: probability of occurrence of the remaining sample.

- Gaussian distribution: distribution of a population when N, the population size, is a very large value

$$W_G = \frac{1}{\sigma\sqrt{2 \cdot \pi \cdot}} \cdot \exp\left(\frac{-(n_1 - \overline{n_1})^2}{2\sigma^2}\right),$$

where σ is the standard deviation.

Happy holidays!

Credit: netfolk.blog.hu

The End

Thank you for your attention!

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